

MATHEMATICS FOR COMPUTING

Linear Algebra

Vectors, Matrices, Linear Transformations
and Vector Spaces

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2026

| Topics Covered

1. Vectors and coordinate systems
2. Vector addition and scalar multiplication
3. Linear combinations, span and basis vectors
4. Matrix multiplication as composition
5. Three-dimensional linear transformations
6. The determinant
7. Inverse matrices
8. Column space and null space
9. Nonsquare matrices as transformations between dimensions
10. Dot products and duality
11. Cross products
12. Cross products in the light of linear transformations
13. Cramer's rule
14. Change of basis
15. Eigenvectors and eigenvalues
16. Vector spaces

The presentation contains 55 slides, with worked examples and revision points for these topics.

| Vectors: The Basic Object

A vector represents a quantity with both **magnitude** and **direction**. In coordinate form, it is represented by an ordered list of components.

- A 2D vector has two components.
- A 3D vector has three components.
- The components describe displacement along coordinate directions.
- Vectors are central to geometry, transformations and data representation.

$$\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

| Vector Notation and Dimension

For a vector in $\mathbb{R}^3$,

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}.$$

The entries are its **components**. The number of components determines the dimension of the coordinate representation.

- $\mathbb{R}^2$: ordered pairs (x, y) .
- $\mathbb{R}^3$: ordered triples (x, y, z) .
- A vector may be written as a column vector or using unit-vector notation.

The notation distinguishes the object being represented from the particular coordinates used to represent it.

| Magnitude of a Vector

The magnitude gives the length of a vector.

$$\|\mathbf{v}\| = \sqrt{x^2 + y^2 + z^2}$$

For $\mathbf{v} = (3, 4)$,

$$\|\mathbf{v}\| = \sqrt{3^2 + 4^2} = 5.$$

- Magnitude is always non-negative.
- A zero vector has magnitude 0.
- Magnitude is useful when measuring distance, length and scale.

| Coordinate Systems and Unit Vectors

A coordinate system supplies reference directions for describing points and vectors. For three-dimensional space, the standard unit vectors are

$$\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{k}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Hence any vector can be decomposed as

$$\mathbf{v} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

| Vector Addition

Vector addition is performed component by component.

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

Then

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix}$$

- Addition is defined when vectors have the same dimension.
- Geometrically, the operation combines displacements.

| Vector Addition: Properties

For vectors $\mathbf{u}$ and $\mathbf{v}$, addition is component-wise.

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

and

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w}).$$

- $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
- Every vector has an additive inverse $-\mathbf{u}$.
- Geometrically, addition can be visualized using the head-to-tail rule.

Vector Addition: Worked Example

Let

$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix}.$$

Add corresponding components:

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 2 + 4 \\ -1 + 3 \\ 3 - 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}.$$

Result: the sum remains a vector in $\mathbb{R}^3$.

| Scalar Multiplication

A scalar is an ordinary number that changes every component of a vector by the same factor.

$$k \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} kx \\ ky \\ kz \end{bmatrix}$$

For example,

$$-2 \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \\ -4 \end{bmatrix}.$$

- $|k| > 1$ changes the length by a factor greater than one.
- $0 < |k| < 1$ reduces the length.
- A negative scalar reverses direction.

| Effect of Scalar Multiplication

Suppose $\mathbf{v}$ is non-zero.

- Multiplication by 2 produces $2\mathbf{v}$: same direction, twice the length.
- Multiplication by $1/2$ produces half the length.
- Multiplication by -1 reverses the direction.
- Multiplication by 0 produces the zero vector.

Key idea: scalar multiplication changes the scale of a vector without changing its coordinate dimension.

| Linear Combinations

A vector can be constructed by scaling several vectors and adding the results.

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$$

Here $c_1, \dots, c_n$ are scalar coefficients.

- Linear combinations are the building blocks of span and basis.
- Different coefficient choices may produce different vectors.
- The same vector may have more than one representation when the generating vectors are dependent.

| Linear Combination: Example

Take

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Find $3\mathbf{v}_1 - 2\mathbf{v}_2$:

$$3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} - \begin{bmatrix} 4 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \end{bmatrix}.$$

Thus the resulting vector is generated from $\mathbf{v}_1$ and $\mathbf{v}_2$.

| Span of Vectors

The **span** of a set of vectors is the collection of every vector obtainable from their linear combinations.

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}.$$

For the standard directions in $\mathbb{R}^2$,

$$\text{Span}\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\} = \mathbb{R}^2.$$

- Span describes what a set of vectors can generate.
- It is a subspace of the ambient vector space.

| Span: How to Test Generation

Suppose we want to represent

$$\mathbf{x} = \begin{bmatrix} a \\ b \end{bmatrix}$$

using $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$.

$$a\mathbf{e}_1 + b\mathbf{e}_2 = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}.$$

Therefore every vector in $\mathbb{R}^2$ is generated by these two directions. **Conclusion:** their span is all of $\mathbb{R}^2$.

| Basis: Two Conditions

A set of vectors forms a **basis** when it satisfies both conditions:

1. The vectors **span** the space.
2. The vectors are **linearly independent**.

For $\mathbb{R}^3$, the standard basis is

$$\mathbf{e}_1 = (1, 0, 0), \quad \mathbf{e}_2 = (0, 1, 0), \quad \mathbf{e}_3 = (0, 0, 1).$$

A basis gives enough directions to generate the space without unnecessary redundancy.

| Basis Representation

Using the standard basis of $\mathbb{R}^3$,

$$\begin{bmatrix} 4 \\ -2 \\ 5 \end{bmatrix} = 4\mathbf{e}_1 - 2\mathbf{e}_2 + 5\mathbf{e}_3.$$

The coefficients $(4, -2, 5)$ are the coordinates of the vector relative to that basis.

- A basis provides a coordinate system for a vector space.
- Linear independence ensures the representation is unique.

| Matrices: Rectangular Arrays

A matrix is a rectangular arrangement of numbers.

$$A = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 3 & 5 \end{bmatrix}.$$

This matrix has 2 rows and 3 columns, so its order is 2×3 .

- Matrices can encode systems of equations.
- They provide compact representations of linear transformations.
- They are used for rotations, scaling and reflections.

| Matrix Dimensions and Compatibility

If A has m rows and n columns, write $A \in \mathbb{R}^{m \times n}$. For multiplication AB to be defined,

$$A_{m \times n} B_{n \times p}$$

produces an $m \times p$ matrix.

- The inner dimensions must match.
- The outer dimensions give the size of the result.

$$(m \times n)(n \times p) = m \times p$$

| Matrix Multiplication

Matrix multiplication follows the **row-by-column** rule. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1(2) + 2(4) & 1(1) + 2(3) \\ 3(2) + 0(4) & 3(1) + 0(3) \end{bmatrix}.$$

| Matrix Multiplication: Result

Continuing the previous calculation,

$$AB = \begin{bmatrix} 10 & 7 \\ 6 & 3 \end{bmatrix}.$$

Each entry is obtained by taking one row from A and one column from B . **Example of one entry:**

$$(AB)_{12} = 1(1) + 2(3) = 7.$$

Matrix multiplication is not ordinary element-by-element multiplication.

| A Key Property of Matrix Multiplication

Matrix multiplication is generally **not commutative**.

$$AB \neq BA$$

in general. Other useful properties include

$$A(B + C) = AB + AC, \quad A(BC) = (AB)C.$$

- Associativity holds when the products are defined.
- The order of factors matters.
- This becomes important when matrices represent successive transformations.

| Matrix Multiplication: Dimension Check

Suppose

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{n \times p}.$$

Then the product AB exists and

$$AB \in \mathbb{R}^{m \times p}.$$

- The inner dimensions must agree.
- The resulting matrix takes the outer dimensions.
- Each entry is obtained from a row of A and a column of B .

| Matrix Multiplication as Composition

A matrix can represent a transformation of a vector. Suppose

$$T(\mathbf{x}) = A\mathbf{x}, \quad S(\mathbf{x}) = B\mathbf{x}.$$

Applying T first and then S gives

$$S(T(\mathbf{x})) = B(A\mathbf{x}) = (BA)\mathbf{x}.$$

Thus,

$$\boxed{S \circ T \longleftrightarrow BA}$$

| Composition: Order Matters

Consider the sequence

$$\mathbf{x} \xrightarrow{A} A\mathbf{x} \xrightarrow{B} B(A\mathbf{x}).$$

The combined matrix is BA .

- The matrix nearest the vector acts first.
- Reversing the order generally changes the result.
- This connects matrix algebra with sequential geometric operations.

Memory aid: read a composition from right to left.

| Linear Transformations

A transformation T is linear when it preserves vector addition and scalar multiplication:

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}),$$

$$T(c\mathbf{u}) = cT(\mathbf{u}).$$

Every matrix A defines a linear transformation

$$T(\mathbf{x}) = A\mathbf{x}.$$

Linearity means that the structure of vector addition and scaling is preserved.

| Linear Transformation: Example

Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

For $\mathbf{x} = (x, y)$,

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 2x \\ 3y \end{bmatrix}.$$

The transformation stretches the two coordinate directions by different factors.

Observation: the origin remains fixed because $T(\mathbf{0}) = \mathbf{0}$.

| Composition: Order Matters

If

$$T(\mathbf{x}) = A\mathbf{x}, \quad S(\mathbf{x}) = B\mathbf{x},$$

then applying T first and S second gives

$$S(T(\mathbf{x})) = B(A\mathbf{x}) = (BA)\mathbf{x}.$$

Thus

$$S \circ T \longleftrightarrow BA.$$

Important: the matrix on the right acts first.

Three-Dimensional Linear Transformations

A 3×3 matrix acts on vectors in $\mathbb{R}^3$.

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

The output is

$$T(\mathbf{x}) = A\mathbf{x}.$$

- Three coordinates enter the transformation.
- Three coordinates are produced.
- The matrix controls how directions are mixed or scaled.

Three-Dimensional Transformation: Example

Let

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}.$$

Then

$$A\mathbf{x} = \begin{bmatrix} 2 + 1 \\ 6 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 1 \end{bmatrix}.$$

So the matrix maps one point/vector in $\mathbb{R}^3$ to another.

| Determinant of a 2×2 Matrix

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det(A) = ad - bc$$

The determinant is a scalar associated with a square matrix.

- It helps determine whether a square matrix is invertible.
- It also describes geometric scaling under the corresponding transformation.

| Determinant: Worked Example

Consider

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

Then

$$\det(A) = 4(3) - 1(2) = 12 - 2 = 10.$$

Since

$$\det(A) = 10 \neq 0,$$

the matrix has an inverse. **Conclusion:** a non-zero determinant signals invertibility for a square matrix.

| Geometric Meaning of the Determinant

The absolute value of a determinant gives the area-scaling factor in two dimensions.

$$|\det(A)|$$

represents how the area of a region changes under the transformation.

- $|\det(A)| > 1$: area is enlarged.
- $0 < |\det(A)| < 1$: area is reduced.
- $\det(A) < 0$: orientation is reversed as well as scaled.

In three dimensions, the analogous interpretation concerns volume scaling.

| Zero Determinant and Invertibility

If

$$\det(A) = 0,$$

the transformation collapses at least one independent direction. Therefore,

$$\boxed{\det(A) = 0 \implies A^{-1} \text{ does not exist.}}$$

- The matrix loses information along some direction.
- Its columns cannot provide the required full set of independent directions.
- The corresponding linear system does not have a unique solution for every right-hand side.

| Determinant: What It Tells Us

For a square matrix, the determinant gives information about the transformation.

- $|\det(A)|$ gives the area-scaling factor in 2D.
- In 3D, $|\det(A)|$ gives the volume-scaling factor.
- A negative determinant also indicates an orientation reversal.
- If $\det(A) = 0$, the transformation collapses dimensions and the matrix is not invertible.

| Inverse Matrix

The inverse of A is denoted by A^{-1} and satisfies

$$AA^{-1} = A^{-1}A = I.$$

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

provided $ad - bc \neq 0$.

| Inverse Matrix: Example

Take

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}.$$

First,

$$\det(A) = 3(1) - 1(1) = 2.$$

Hence

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}.$$

A quick verification is obtained by multiplying A and A^{-1} to obtain I .

| Inverse Matrix: Existence and Meaning

The inverse A^{-1} reverses the action of A when it exists:

$$AA^{-1} = A^{-1}A = I.$$

For a 2×2 matrix,

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Therefore,

$$\det(A) \neq 0 \iff A^{-1} \text{ exists.}$$

| Column Space

The **column space** of a matrix is the span of its columns. For

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

its columns are

$$\mathbf{c}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2\mathbf{c}_1.$$

Therefore,

$$\text{Col}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}.$$

| Meaning of Column Space

The column space describes the set of all possible outputs of $A\mathbf{x}$.

- If the columns are independent, each contributes a new direction.
- Dependent columns do not enlarge the span.
- Column space is closely connected with the range of a linear transformation.

$$\text{Col}(A) = \{A\mathbf{x} : \mathbf{x} \text{ is an allowed input}\}$$

| Column Space: Interpretation

The column space is the span of the columns of A :

$$\text{Col}(A) = \text{Span}\{\text{columns of } A\}.$$

It represents the set of all vectors that can appear as outputs $A\mathbf{x}$.

- It is a subspace of the output space.
- Independent columns determine independent output directions.
- Column-space dimension is the rank of the matrix.

| Null Space

The **null space** consists of all vectors mapped to the zero vector:

$$\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$$

For

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

let $\mathbf{x} = (x, y)^T$. Then

$$x + 2y = 0.$$

So the null space contains all vectors satisfying $x = -2y$.

| Null Space: Parameter Form

Set $y = t$. Then

$$x = -2t.$$

Therefore

$$\mathbf{x} = \begin{bmatrix} -2t \\ t \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Hence

$$\text{Null}(A) = \text{Span} \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$$

A parameter form makes every solution visible at once.

| Null Space: Interpretation

The null space contains every vector satisfying

$$A\mathbf{x} = \mathbf{0}.$$

It describes input directions that are mapped to the zero vector.

- The zero vector is always in the null space.
- A non-zero null-space vector indicates loss of information under the transformation.
- Null space is a subspace of the input space.

| Nonsquare Matrices as Transformations

A nonsquare matrix can map between spaces of different dimensions. Consider

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}.$$

This is a 2×3 matrix, so it defines

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^2.$$

- Three input coordinates are accepted.
- Two output coordinates are produced.

| Nonsquare Matrix: Example

For

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

we obtain

$$A\mathbf{x} = \begin{bmatrix} x + 2z \\ y + 3z \end{bmatrix}.$$

Thus a 3D input is represented by a 2D output. **Key point:** the matrix dimensions tell us the input and output dimensions of the transformation.

| Dot Product

For vectors

$$\mathbf{u} = (u_1, u_2, u_3), \quad \mathbf{v} = (v_1, v_2, v_3),$$

the dot product is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3$$

It can also be expressed geometrically as

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

The result of a dot product is a **scalar**, not a vector.

| Dot Product and Orthogonality

Two non-zero vectors are orthogonal when their dot product is zero:

$$\boxed{\mathbf{u} \cdot \mathbf{v} = 0 \implies \mathbf{u} \perp \mathbf{v}}$$

Example:

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix} = 2 - 2 = 0.$$

Therefore the two vectors are perpendicular. The dot product provides an algebraic test for geometric perpendicularity.

| Duality and Linear Functionals

Fix a vector $\mathbf{a}$. The rule

$$f(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x}$$

maps vectors to real numbers. For

$$\mathbf{a} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad f \begin{pmatrix} x \\ y \end{pmatrix} = 2x + 3y.$$

This is a **linear functional**. **Idea:** duality connects vectors with linear functions that take vectors as inputs and return scalars.

| Dot Product: Orthogonality

For two non-zero vectors,

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

If

$$\mathbf{u} \cdot \mathbf{v} = 0,$$

then $\cos \theta = 0$ and the vectors are perpendicular. **Therefore:** the dot product provides an algebraic test for orthogonality.

| Cross Product

The cross product is defined for vectors in $\mathbb{R}^3$ and produces a vector perpendicular to both inputs. For

$$\mathbf{u} = (u_1, u_2, u_3), \quad \mathbf{v} = (v_1, v_2, v_3),$$

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}$$

The order matters: generally $\mathbf{u} \times \mathbf{v} \neq \mathbf{v} \times \mathbf{u}$.

Cross Product: Direction and Magnitude

For vectors in $\mathbb{R}^3$,

$$\mathbf{u} \times \mathbf{v}$$

is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$. Its magnitude is

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta.$$

Hence it is related to the area of the parallelogram spanned by the two vectors.

Cross Product: Example

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

Then

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 0(0) - 2(1) \\ 2(2) - 1(0) \\ 1(1) - 0(2) \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}.$$

The result is perpendicular to both input vectors.

| Geometric Meaning of the Cross Product

The magnitude is

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$$

This is the area of the parallelogram formed by the two vectors. Therefore the corresponding triangle has area

$$\frac{1}{2} \|\mathbf{u} \times \mathbf{v}\|.$$

If the vectors are parallel, $\theta = 0$ and the cross product is the zero vector.

Cross Product as a Linear Transformation

Fix

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

and define

$$T(\mathbf{x}) = \mathbf{a} \times \mathbf{x}.$$

This can be written using the matrix

$$T(\mathbf{x}) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \mathbf{x}.$$

Hence cross product with a fixed vector is a linear transformation.

| Cramer's Rule

Cramer's rule provides a determinant-based method for solving a system of linear equations. For

$$ax + by = e, \quad cx + dy = f,$$

the coefficient determinant is

$$D = ad - bc.$$

When $D \neq 0$, the solution is unique and

$$x = \frac{ed - bf}{D}, \quad y = \frac{af - ec}{D}.$$

| Cramer's Rule: Example

Solve

$$2x + y = 7, \quad x - y = 1.$$

The coefficient determinant is

$$D = 2(-1) - 1(1) = -3.$$

Using elimination or Cramer's rule gives

$$x = \frac{7(-1) - 1(1)}{-3} = \frac{-8}{-3} = \frac{8}{3}.$$

Then

$$y = x - 1 = \frac{5}{3}.$$

Cramer's rule is especially direct for small systems, although determinant calculations grow for larger systems.

| Change of Basis

The coordinates assigned to a vector depend on the chosen basis. Suppose B and C are two bases. Then

$$[\mathbf{v}]_C = P_{C \leftarrow B} [\mathbf{v}]_B$$

where $P_{C \leftarrow B}$ is the change-of-basis matrix.

- The underlying vector does not change.
- Only its coordinate description changes.
- Different bases can make the same vector easier to represent in different contexts.

| Cramer's Rule: When It Applies

For a square system

$$A\mathbf{x} = \mathbf{b},$$

Cramer's rule uses determinants to obtain individual unknowns. The key condition is

$$\det(A) \neq 0.$$

Then the coefficient matrix is invertible and the system has a unique solution.

Cramer's rule is particularly useful for small systems where determinant calculations are manageable.

| Change of Basis: Example

Let

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

and suppose

$$[\mathbf{v}]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Then the actual vector is

$$\mathbf{v} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}.$$

Thus its standard coordinates are $[\mathbf{v}]_E = (5, -1)^T$.

| Change of Basis: Why It Matters

A vector itself does not change when the basis changes; its coordinate column changes.
If B and C are two bases,

$$[\mathbf{v}]_C = P_{C \leftarrow B} [\mathbf{v}]_B.$$

- The change-of-basis matrix converts coordinates from one basis to another.
- Different bases can make the same vector easier to represent or analyze.

| Eigenvalues and Eigenvectors

For a square matrix A , an eigenvector satisfies

$$A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \neq \mathbf{0}$$

where λ is the corresponding eigenvalue. Rearranging,

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

A non-zero solution exists when

$$\det(A - \lambda I) = 0$$

which is called the characteristic equation.

| Eigenvalues: Characteristic Equation

An eigenvalue satisfies

$$A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \neq \mathbf{0}.$$

Rearranging,

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

For a non-zero solution to exist,

$$\det(A - \lambda I) = 0.$$

This equation is called the **characteristic equation**.

| Eigenvalues: Worked Example

Consider

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}.$$

Then

$$\det(A - \lambda I) = (4 - \lambda)(2 - \lambda).$$

Setting this equal to zero gives

$$\lambda_1 = 4, \quad \lambda_2 = 2.$$

For $\lambda = 4$,

$$(A - 4I)\mathbf{v} = 0$$

leads to an eigenvector in the direction of $(1, 0)^T$.

| Eigenvectors: Finding the Direction

After obtaining an eigenvalue λ , substitute it into

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

The non-zero solutions form the eigenspace associated with λ .

- Any non-zero scalar multiple of an eigenvector is also an eigenvector for the same eigenvalue.
- Eigenvectors corresponding to different eigenvalues provide distinct invariant directions under standard conditions.

| Eigenvectors: Geometric Meaning

For a general vector, applying A can change both its direction and magnitude. For an eigenvector,

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Therefore its direction is preserved while its magnitude is scaled by $|\lambda|$.

- **Eigenvector:** a special invariant direction.
- **Eigenvalue:** the scaling factor associated with that direction.
- Eigenvectors help reveal important structural directions of a matrix transformation.

| Vector Spaces: Examples and Dimension

A vector space is closed under vector addition and scalar multiplication and satisfies the vector-space axioms. Examples include

$\mathbb{R}^2$, $\mathbb{R}^3$, P_2 , matrix spaces and function spaces.

The dimension equals the number of vectors in a basis. Thus

$$\dim(\mathbb{R}^2) = 2, \quad \dim(\mathbb{R}^3) = 3,$$

and for P_2 , the basis $\{1, x, x^2\}$ gives dimension 3.

| Vector Spaces and Dimension

A vector space is a collection of objects closed under vector addition and scalar multiplication and satisfying the vector-space axioms. Examples include

- $\mathbb{R}^2$ and $\mathbb{R}^3$,
- polynomial spaces,
- matrix spaces,
- function spaces.

The **dimension** is the number of vectors in a basis.

$$\dim(\mathbb{R}^2) = 2, \quad \dim(\mathbb{R}^3) = 3.$$

For $P_2 = \{a + bx + cx^2\}$, the basis $\{1, x, x^2\}$ gives $\dim(P_2) = 3$.

| Important Relationships and Quick Revision

Concept flow

Vectors $\rightarrow$ Linear Combinations $\rightarrow$ Span $\rightarrow$ Basis $\rightarrow$ Dimension

Matrix $\rightarrow$ Linear Transformation $\rightarrow$ Determinant / Inverse / Spaces

Essential formulas

- $\|\mathbf{v}\| = \sqrt{x^2 + y^2 + z^2}$
- $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$
- $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$
- $\det(A) = ad - bc$ for 2×2 matrices
- $A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
- $A\mathbf{v} = \lambda\mathbf{v}$ for eigenvectors

Thank you